Instructional Designer, Work Sample

Narrative Script

Topic: The Condenser | Subtopic: Parts and Functions

Instructional Designer: Rayene Medabis

Learning Objective: *By the end of this lesson, the learner will be able to identify the key components of a condenser unit, describe what each one does in the cooling process, and locate the outdoor disconnect for safe maintenance.*

Slide	Narration	Visual Callout	Notes for Designer
1	Ever looked at that big metal box outside a building and wondered what it actually does? That's the condenser unit. Today, we're going to open it up , piece by piece , so you understand exactly how it works, and where to go if you ever need to shut it off safely.	Wide establishing shot of an outdoor condenser unit beside a house. A soft pulsing glow draws the eye to the unit , no labels yet.	Keep this clean and uncluttered , the goal is curiosity, not information. Simple flat-color illustration or photo cutout, not a diagram.
2	This whole unit is called the condenser. Its one job is to take the heat pulled out of the building and release it into the outdoor air. Every part inside , the fan, the coil, the electrical components , works together to make that happen.	Simple 2D cutaway of the condenser showing empty outlines for each internal part (unlabeled), with a faint heat arrow moving from inside the building to outside.	This is the 'map' shot the learner will return to later. Flat vector style, muted palette , save bright color for the parts as we introduce them.
3	First, the housing. This metal cabinet protects everything inside from rain, dirt, and damage, while still allowing air to flow freely through the sides.	A simple shield icon or animated rain droplets bounce off the panel to show protection.	Quick, light animation , 2 seconds max. This slide should feel like the shortest, simplest beat in the sequence.
4	Inside, the fan pulls outside air across the coil and pushes the collected heat out into the atmosphere. Without airflow, that heat would have nowhere to go.	Top-down flat illustration of the fan blades turning slowly, with directional air arrows flowing from outside, through the unit, and back out.	Loop the fan rotation at a readable speed (not realistic RPM). Air arrows should be in the same direction the narration describes.
5	Wrapped around the inside is the coil , a set of metal tubes carrying hot refrigerant from inside the building. As air from the fan moves across it, heat transfers out of the refrigerant and into the outside air. That's the moment cooling actually happens.	Looping coil illustrated with a color gradient , refrigerant shown red/hot entering, fading to orange/cooler as it exits. Small heat-particle icons lift away into the passing air.	Color-code the temperature change so it reads instantly (red to orange).
6	The contactor is an electrical switch. When the thermostat calls for cooling, the contactor closes and sends power to the fan and	Simple switch icon flips from open to closed. A short pulse of light travels from the switch out to icons representing the fan and	Snap the switch closed crisply , this should feel like a deliberate 'click' moment, both visually and

Slide	Narration	Visual Callout	Notes for Designer
	compressor, starting the whole cycle.	compressor.	in narration pacing.
7	Right next to it is the capacitor. Think of it as a small battery that gives the fan motor an extra boost of energy to get started, then helps keep it running smoothly.	Capacitor icon with a small upward energy-spike animation feeding into the fan motor icon.	Keep the 'battery boost' metaphor visual and simple , one spike, one arrow. Avoid technical waveform or voltage diagrams.
8	The defrost board acts as the brain of the unit. It monitors temperature and tells the system when to pause briefly and melt off any ice that builds up on the coil.	Small circuit-board icon with a gentle 'thinking' pulse, followed by a quick before/after: frost forming on the coil, then clearing away.	Two-part beat: frost building (1–2 sec), board pulses, frost clears (1–2 sec). Keep it snappy so it doesn't slow the pacing.
9	Before doing any maintenance, always locate the outdoor disconnect first , usually a small gray box mounted on the wall near the unit. Pulling it out cuts power to the entire condenser, so you can work safely without risk of electrical shock.	Wide shot showing the disconnect box mounted beside the condenser. A hand icon pulls the disconnect block out; a power icon fades from red (on) to gray (off).	Safety-critical moment , make the on/off color change unmistakable (red to gray). This is the one visual the learner most needs to remember.
10	So together: the contactor and capacitor get things started, the fan pulls air across the coil, the coil releases the heat, and the defrost board keeps the whole system running clean. That's the condenser , several small parts, working as one system.	The full cutaway diagram from Slide 2 reassembles, with each part briefly highlighting in sequence as it's mentioned: contactor, capacitor, fan, coil, defrost board.	This is the payoff shot. Sync each highlight pop to the matching word in the narration , brisk pacing, no more than half a second per highlight.
11	Next time you see a condenser unit, you'll know exactly what's happening inside it , and just as importantly, exactly where to go first to shut it off safely.	Return to the wide shot from Slide 1, now with every part faintly labeled. The outdoor disconnect glows once more for emphasis before the frame settles.	Bookend with Slide 1's shot for a sense of closure. Let the disconnect glow linger half a second longer than the other labels , final safety reinforcement.

Instructional Designer, Work Sample

Slide-by-Slide Design Rationale

Topic: The Condenser | Subtopic: Parts and Functions

Instructional Designer: Rayene Medabis

Purpose of This Document

This document walks through the Condenser Components storyboard slide by slide. For each slide it explains what the learner is seeing, why that specific visual was chosen over alternatives, how the animation or visual treatment communicates the underlying system behavior or component function, and the instructional-design thinking behind its placement in the sequence. Where useful, it draws on the narration and "Notes for Designer" column from the Deliverable 1 narrative script to show how narration, visual, and pacing decisions were made together rather than independently.

The lesson's stated objective is for the learner to identify the key components of a condenser unit, describe what each one does in the cooling process, and locate the outdoor disconnect for safe maintenance. Every design decision below is traceable back to one of those three outcomes.

Overall Design Framework

Six recurring design principles shape the storyboard as a whole, and are referenced throughout the slide-by-slide breakdown:

Concrete before abstract — The lesson opens on a photorealistic backyard scene of an actual unit, then moves to a flat vector cutaway, then to individual component icons, and only then to abstract electrical concepts (switches, energy pulses, control logic). Familiar, tangible imagery is established before symbolic or diagrammatic imagery is introduced.

The recurring "system map" — The same cutaway diagram is introduced once (Slide 3) as an unlabeled map, then reused, fully labeled, as the recap payoff (Slide 12), and again as the practice exercise (Slides 15–16). Reusing one base image rather than generating a new diagram each time turns the diagram itself into a memory anchor the learner returns to and progressively decodes.

Motion and color stand in for invisible physics — Heat, airflow, and electrical current cannot be seen in real life. Each is made visible through a consistent visual convention: directional arrows for airflow, a red-to-orange gradient for heat loss in the coil, and light pulses for electrical current. Once a convention is established it is reused rather than replaced, so the learner doesn't have to relearn a new visual language each slide.

Deliberate phase dividers — "Revise" and "Practice" title cards mark clear breaks between teach, review, and test phases, using the same establishing photograph each time so the learner recognizes the pattern instantly rather than having to reorient.

Weighted repetition for the safety-critical item — The outdoor disconnect is the only piece of content that appears four separate times (Slides 2, 10, 13, 15–16) and is the only element treated in warning red. This is intentional redundancy reserved for the one action with real physical consequences if forgotten.

Retrieval before feedback — The practice section asks the learner to label the diagram from memory (Slide 15) before showing the answer key (Slide 16), rather than presenting the labeled version first — active recall is required before feedback is given.

Phase 1: Hook & Framing

Slides 1–2 establish the topic, state the objective up front, and create curiosity before any teaching begins.

Slide 1 — Title Slide — "The Condenser"

What the learner sees: A detailed, realistic illustration of a full condenser system (indoor components and the outdoor unit together), the title "The Condenser," the subtitle "Parts and Functions," the full learning objective written out in plain language, and an authorship credit.

Why this visual was chosen: A richly rendered, almost photographic illustration is used here instead of a diagram or icon set. This is the only point in the lesson where the whole real-world system is shown in this level of visual detail, which sets a baseline: "this is what the actual thing looks like." Every diagram that follows is a simplified abstraction of this image, so starting with the realistic version gives the learner something concrete to mentally map the simplified visuals back onto.

How it supports system understanding: Because nothing is labeled or explained yet, the image communicates scale and complexity rather than function — it signals "there's a lot going on here that we're about to make simple," which is the emotional set-up for the curiosity hook on the next slide.

Instructional thinking: Stating the objective in full on the very first slide is a deliberate transparency move: the learner knows exactly what they'll be able to do by the end before any content is delivered, which helps them self-monitor progress through the rest of the lesson.

Slide 2 — Establishing Shot

What the learner sees: A painted backyard scene with a condenser unit sitting beside a house, exactly like one a learner might walk past in real life. A soft orange circle and arrow draw the eye to the unit, and a caption reads that this is the metal box most people walk past without knowing what it does.

Why this visual was chosen: The narrative script's designer notes are explicit here: "keep this clean and uncluttered, the goal is curiosity, not information," and specify a simple flat-color illustration or photo cutout rather than a diagram. No labels, no callouts beyond the single caption — the slide is intentionally under-informative.

How it supports system understanding: By withholding detail, the visual creates a curiosity gap: the learner can see the object but not yet understand it. That gap is what the rest of the lesson exists to close, so the slide is doing rhetorical work (motivation) rather than teaching work (information transfer).

Instructional thinking: Opening with a scene the learner has plausibly experienced firsthand (rather than a technical diagram) builds relevance before competence — a common early move in instructional sequencing, since learners engage more readily with content once they see why it matters to them personally.

Phase 2: Building the Mental Model

Slide 3 gives the learner a single system-level idea — heat rejection — before any individual component is introduced, so later detail has somewhere to attach.

Slide 3 — "One Job: Release the Heat"

What the learner sees: A cutaway diagram of the condenser showing the fan, coils, compressor, and refrigerant lines, with a heat arrow flowing from inside the building wall out through the unit. Text states the condenser's single job, and a caption reframes the unit as the "exhale" part of the cooling system.

Why this visual was chosen: This slide functions as an advance organizer: instead of starting with the first component in isolation, it shows the whole system performing its one job (moving heat outdoors) before any part is explained individually. The narrative script's designer notes flag this explicitly as "the 'map' shot the learner will return to later," confirming it was built to be reused, not just viewed once.

How it supports system understanding: The heat arrow turns an abstract statement ("the condenser releases heat") into a visible path with a direction and an endpoint. The "exhale" analogy does the same work in language — it maps an unfamiliar mechanical process onto a bodily process every learner already understands, giving the concept an intuitive handle before any technical vocabulary is introduced.

Instructional thinking: Placing this single-function overview immediately after the hook, and before any of the six components, means every subsequent component slide (Housing, Fan, Coil, Contactor, Capacitor, Defrost Board) can be taught as "one piece of the machine you already understand the purpose of," rather than as a standalone fact. This is what makes the recap on Slide 12 — the same diagram, now fully labeled — land as a payoff rather than a repeat: the learner is completing a picture they were shown empty at the start.

Phase 3: Component Walkthrough

Slides 4–9 teach the six components one at a time, in an order that follows the physical cause-and-effect sequence of the machine rather than an arbitrary list.

Slide 4 — The Housing

What the learner sees: A close-up of the unit's louvered metal panel with rain droplets bouncing off it and a small shield icon, alongside text explaining that the housing protects internal parts while still letting air flow through.

Why this visual was chosen: The designer notes describe this as intentionally "the shortest, simplest beat in the sequence," with the animation capped at two seconds. Housing is a purely structural, non-moving part, so it was chosen as the entry point into the component walkthrough — the easiest possible concept to open on.

How it supports system understanding: The rain-droplet animation makes an invisible function (protection) visible by showing what the housing prevents, rather than only describing it in text. Watching water deflect off the surface requires no prior technical knowledge to interpret correctly.

Instructional thinking: Starting the six-part walkthrough with its simplest, purely structural member is a scaffolding choice: it lets the learner succeed at understanding one easy component before the lesson asks them to track moving air, heat transfer, and electrical switching in the slides that follow.

Slide 5 — The Fan

What the learner sees: A cutaway showing two fan motors, directional arrows for outside air coming in and hot air going out, and labels for the condenser coils, compressor, and refrigerant lines.

Why this visual was chosen: The fan is the first moving, functional component in the sequence, and the visual exists to make airflow — which is invisible in real life — visible and directional.

How it supports system understanding: The designer notes specify looping the fan rotation "at a readable speed (not realistic RPM)." A real condenser fan spins too fast for the blades to be individually tracked; slowing the animation down sacrifices realism in favor of legibility, so the learner can actually watch the blade motion and connect it to the air arrows rather than seeing a blur. The arrows are also choreographed to match the narration's description of airflow direction, so the audio and visual channels reinforce the same information at the same moment.

Instructional thinking: Fan is taught immediately after Housing and before Coil because airflow is the physical precondition for the coil's heat exchange — without moving air, the coil in the next slide has nothing to transfer heat into. Sequencing the components in their true causal order (air moves, then heat transfers) lets the lesson's structure mirror the machine's own logic, rather than an arbitrary component list.

Slide 6 — The Coil

What the learner sees: A 3D-style cutaway of the coil with refrigerant shown in a red-to-orange color gradient as it moves through the tubing, small particle icons lifting away from the metal to represent heat leaving, and labels marking where hot vapor enters and cooler liquid exits.

Why this visual was chosen: This slide is the conceptual centerpiece of the whole lesson — it's where the narration explicitly says "cooling actually happens." It follows the fan slide directly so that the air the learner just watched moving now has an obvious job to do.

How it supports system understanding: The design notes call for color-coding the temperature change so it "reads instantly" — red for hot refrigerant entering, fading to a cooler tone as it exits. This lets the learner see the transfer of heat as a color change rather than a technical description, which is far faster to process and doesn't require any prior understanding of refrigerant or thermodynamics. The lifting particle icons give the abstract idea of "heat leaving the system" a physical, animated form the eye can follow.

Instructional thinking: Coil is deliberately the sixth slide overall but only the second component taught, right after Fan — this placement ensures the learner already has "moving air" established as a fact before being asked to understand what that air does when it crosses the coil. Teaching heat transfer before airflow was in place would have made the coil's function harder to justify visually.

Slide 7 — The Contactor

What the learner sees: A switch icon flipping from open to closed, a thermostat showing a 72°F call for cooling, and a pulse of light traveling from the switch out to icons representing the fan and compressor.

Why this visual was chosen: This slide marks a deliberate shift from the mechanical/thermal half of the lesson (housing, fan, coil) into the electrical/control half (contactor, capacitor, defrost board). A binary, discrete switch animation was chosen specifically because a contactor's behavior is genuinely binary — it is either open or closed — unlike the continuous processes of airflow and heat transfer shown in the previous two slides.

How it supports system understanding: The designer notes call for the switch to "snap... closed crisply," describing it as a deliberate 'click' moment in both the visual and the narration pacing. That crispness is itself communicative: a sharp, sudden animation tells the learner this is an instantaneous triggered event, not a gradual process, which is an accurate representation of how a contactor actually behaves.

Instructional thinking: Placing Contactor right after Coil and before Capacitor groups the two electrical components together and sets up the pairing explained on the next slide — the contactor turns power on, and the capacitor (next) is what gets the motor actually moving once power is available.

Slide 8 — The Capacitor

What the learner sees: A capacitor icon with a small energy-spike arrow feeding into a fan motor icon, described in text as a battery-like boost that starts the motor and then helps it run smoothly.

Why this visual was chosen: The designer notes are explicit that the "battery boost" metaphor should stay visual and simple — one spike, one arrow — and specifically avoid a technical waveform or voltage diagram. This is a case where more technically accurate visualization (an oscilloscope-style AC waveform) was deliberately rejected in favor of a simpler, if less precise, analogy that matches the audience's needs.

How it supports system understanding: A single spike-and-arrow animation keeps the idea legible at a glance: energy goes in, motor gets moving. Adding technical waveform detail would introduce information the stated learning objective doesn't require and would risk overwhelming a learner who only needs a functional understanding, not an electrical-engineering one.

Instructional thinking: Sequencing Capacitor immediately after Contactor lets the two slides function as one composite idea — "getting the system started is a two-part electrical handoff" — rather than two disconnected facts the learner has to link together on their own.

Slide 9 — The Defrost Board

What the learner sees: A small circuit-board icon described as "the brain of the unit," shown "monitoring" temperature, followed by a before/after sequence of the coil: frost building up, then clearing away.

Why this visual was chosen: This is the last and most abstract of the six components — a feedback-control function with no obvious physical form the way a fan or coil has. It's placed last in the walkthrough deliberately, because its payoff (frost melting off the coil) only makes sense to a learner who already understands what the coil is and what it does, which was taught two slides earlier.

How it supports system understanding: The two-part before/after animation — frost forming, board "pulses," frost clearing — turns an invisible feedback loop into a simple, visible cause-and-effect story. The designer notes call for each half to run only one to two seconds so the beat stays "snappy," keeping momentum going into the safety section that follows rather than letting the lesson's pace drag right before its most important content.

Instructional thinking: Ending the component walkthrough on the Defrost Board completes the electrical/control trio (Contactor, Capacitor, Defrost Board) as a clean group, closing out the "how it works" portion of the lesson before the tone shifts to the safety-critical procedural content in the next slide.

Phase 4: Safety-Critical Procedure

Slide 10 is the highest-priority content in the lesson and is treated with distinct visual language — red warning color, procedural steps, and an explicit call-out as the one thing the learner most needs to remember.

Slide 10 — Safety First: The Disconnect

What the learner sees: The condenser unit next to its gray outdoor disconnect box, a large red arrow banner instructing the learner to locate and use the disconnect for safe maintenance, a three-step "power shut-off procedure" (locate box, open cover, turn off switch/pull plug), an inset showing the switch in the OFF position, and a second red banner instructing the learner to verify power is off with a multimeter before maintenance.

Why this visual was chosen: This slide carries more visual weight — more red, more explicit step numbering, more repeated warning banners — than any other slide in the deck. That's a deliberate departure from the flat, muted palette used everywhere else, and it matches the objective, which names locating the outdoor disconnect as a specific, required outcome rather than general background knowledge.

How it supports system understanding: Red is used here specifically as a safety signal, borrowing a color convention learners already recognize from everyday warning signage, so the danger register is communicated instantly without needing to be read. The numbered shut-off procedure (locate, open, turn off) turns an abstract instruction — "disconnect power safely" — into a concrete, ordered sequence of physical actions, which is what the learner will actually need to execute, not just recognize.

Instructional thinking: This slide comes after all six components have been taught, not before — the learner now understands what they're about to power down and why it matters, which gives the safety instruction meaning it wouldn't have if it were front-loaded before any component context existed. Design notes describe this as "the one visual the learner most needs to remember," which is why it's the first of four total appearances of the disconnect across the lesson (Slides 2, 10, 13, and 15–16) — every other component appears once.

Phase 5: Review & Reinforcement

Slides 11–13 consolidate the lesson before testing begins — first a clear phase break, then a full component recap, then a dedicated second pass on the safety procedure.

Slide 11 — "Revise" (Section Divider)

What the learner sees: The same backyard establishing shot from Slide 2, now overlaid with the single word "Revise."

Why this visual was chosen: This slide introduces no new content — its only job is to mark a phase change from teaching into review. Reusing the opening photograph rather than a new image is deliberate: it bookends the "teach" portion of the lesson and mentally returns the learner to the whole, real-world object after nine slides of increasingly detailed internal diagrams.

How it supports system understanding: By stepping back to the plain establishing shot, the slide resets the learner from "detail mode" (individual parts) to "whole system mode," priming them to reassemble everything they've just learned into one coherent picture on the following slide.

Instructional thinking: An explicit, low-content signpost slide like this reduces cognitive load: rather than the learner having to infer from content alone that the lesson has shifted from new material to review, the slide states it outright, freeing mental effort for the actual recall task ahead.

Slide 12 — "Now You Know:"

What the learner sees: The same base cutaway illustration first shown unlabeled on Slide 3, now fully labeled and numbered: Contactor, Capacitor, Fan, Coil, and Defrost Board, each with a one-line function reminder, highlighted in sequence.

Why this visual was chosen: The narrative script calls this "the payoff shot." Reusing the exact base image from Slide 3, rather than generating a new diagram, is what makes it a payoff rather than a repeat — the learner is watching the same empty map from the start of the lesson get filled in with everything they now know.

How it supports system understanding: The design notes specify syncing each highlight to the matching word in the narration, with no more than half a second per highlight — a brisk, sequential reveal that mirrors the order the components were originally taught in. This lets the learner self-check recall part by part, in the same order they learned it, rather than being shown all five labels at once.

Instructional thinking: Reusing a visual the learner already encoded once (on Slide 3) is a retrieval-practice technique: recognizing the same diagram now fully explained reinforces memory more effectively than a fresh summary image would, because the learner is completing a picture rather than processing new information.

Slide 13 — "Remember!"

What the learner sees: A split-screen slide: the labeled backyard illustration (fan assembly, coil fins, service access panel) beside a dedicated "Safe Shut-Off Procedure" graphic showing a hand pulling the disconnect handle to the OFF position, labeled "POWER OFF (SAFE)."

Why this visual was chosen: This is a second, separate recap slide devoted only to the disconnect procedure — it is deliberately not folded into the five-component recap on Slide 12. Isolating it protects the safety message from being diluted among five other reminders competing for the learner's attention.

How it supports system understanding: Showing a hand physically performing the pull-to-disconnect action (rather than just the box itself) keeps the visual anchored to behavior, not just recognition — it reinforces that this is a physical action to perform, not a fact to recall.

Instructional thinking: Repeating the disconnect procedure here — the third of its four total appearances — is intentional spaced repetition for the single highest-stakes item in the lesson. Placing this repetition close to the end of instruction, just before practice, increases the odds it's still fresh when the learner is tested and, more importantly, when they encounter a real unit later.

Phase 6: Practice & Retrieval

Slides 14–16 shift the learner from recognizing content to producing it unprompted, then give immediate, self-directed feedback.

Slide 14 — "Practice" (Section Divider)

What the learner sees: The same backyard establishing shot used for Slides 2 and 11, now overlaid with the word "Practice."

Why this visual was chosen: This divider mirrors the "Revise" divider structurally — same photo, same treatment, different single word — so the learner instantly recognizes the pattern of a phase change without needing new instructions to understand what kind of content is coming.

How it supports system understanding: No new system content is presented here; the visual's only function is wayfinding within the lesson's structure.

Instructional thinking: Separating "Revise" from "Practice" as two distinct dividers — rather than combining review and testing into one phase — reflects a deliberate choice to first let the learner passively re-encounter the content (recap), and only then require them to actively retrieve it unprompted (practice). Recognition followed by unprompted recall is a stronger sequence for retention than recap alone.

Slide 15 — Practice — Unlabeled Diagram

What the learner sees: The identical cutaway diagram and electrical-compartment inset used in the teaching and recap slides, but every label has been replaced with a blank orange bar in the same position: on the fan, housing, coil, condenser unit, contactor, capacitor, and defrost board.

Why this visual was chosen: This is a label-the-diagram retrieval task — the standard practice format for a components-and-functions objective. Using the exact same diagram the learner has already seen twice (Slide 3 unlabeled, Slide 12 labeled) rather than a new image ensures the exercise tests recall of what was taught, not the learner's ability to interpret an unfamiliar illustration.

How it supports system understanding: Each blank sits in the same physical location on the unit as its real label, requiring the learner to connect a component's name and function back to its actual spatial position on a real condenser — which is precisely the "identify the key components" portion of the learning objective, tested in its most literal form.

Instructional thinking: Presenting the blank version before the answer key forces active retrieval before feedback is given, rather than letting the learner passively review a completed diagram. This ordering is what makes the exercise a genuine practice activity instead of a second recap.

Slide 16 — Practice — Labeled Answer Key

What the learner sees: The same diagram, now fully labeled: Fan, Outdoor Disconnect, Housing, Coil, Condenser Unit, Contactor, Capacitor, and Defrost Board, with captions identifying it as "HVAC System – Outdoor Condenser Unit" and "Electrical Compartment Detail (Inset)."

Why this visual was chosen: Following the blank version immediately with its labeled twin gives the learner instant, self-directed feedback rather than leaving them to guess or wait for an instructor — necessary in a self-paced format with no live facilitator.

How it supports system understanding: Because it is the identical diagram with the blanks now filled, the learner can compare their own answers position by position rather than cross-referencing a separate answer list, minimizing the extra effort needed just to check their work.

Instructional thinking: This slide closes the retrieval-practice loop opened on Slide 15 — testing without feedback has limited learning value, so the answer key is placed immediately after, not on a later slide. It also marks the disconnect's fourth and final appearance in the lesson, completing the deliberate repetition pattern reserved for that one safety-critical element.

Phase 7: Closure

The final slide bookends the lesson visually and restates the objective as a closing self-check.

Slide 17 — "Thank You"

What the learner sees: The same richly rendered, realistic illustration style used on the Slide 1 title card, alongside the full learning objective restated in text, with the six component names — housing, fan, coil, contactor, capacitor, and defrost board — bolded and highlighted in color, plus the author credit.

Why this visual was chosen: Returning to the same illustration style used to open the lesson gives the storyboard a symmetrical structure: it opens and closes on the same visual register, signaling clearly that the lesson has come full circle.

How it supports system understanding: Bolding the six component names within the restated objective functions as a lightweight closing checklist — a learner can scan the six highlighted terms and self-verify they can now define and locate each one, rather than the closing slide being purely decorative.

Instructional thinking: Restating the objective verbatim at the end, rather than a generic closing message, explicitly closes the loop opened on Slide 1: the learner is shown, in the same words they read before any content was delivered, exactly what they now know — giving them a clear, memorable summary of what the lesson accomplished.

Summary: How the Sequence Supports the Objective

Each of the three parts of the stated learning objective is served by a distinct, traceable stretch of the storyboard rather than being addressed all at once:

- "Identify the key components of a condenser unit" is built across Slides 3–9 (introducing each part against the system map) and directly tested in Slides 15–16 (labeling the same diagram from memory).
- "Describe what each one does in the cooling process" is carried by the function-focused narration and visual metaphor on each component slide (Slides 4–9), and reinforced through the sequential recap on Slide 12.
- "Locate the outdoor disconnect for safe maintenance" is the only objective element repeated four separate times (Slides 2, 10, 13, 15–16) and the only content rendered in warning red — a deliberate weighting toward the one outcome with real physical safety consequences.

Taken together, the storyboard moves the learner from a real-world object they recognize but don't understand, through a single-function overview, into a component-by-component build in true physical cause-and-effect order, into safety procedure, and finally through review and tested recall — ending exactly where it began, on the same image, now fully understood.

Instructional Design Framework & Theoretical Approach

Topic: The Condenser | Subtopic: Parts and Functions

Instructional Designer: Rayene Medabis

Purpose of This Document

The narrative script, storyboard, and slide-by-slide design rationale each show what was built and why a specific choice was made on a specific slide. This document steps back one level further: it names the instructional design process and the learning theories that generated those choices in the first place, and shows, model by model, exactly where each one is visible in the finished work. Nothing here is a new design decision, it is the theoretical account of decisions already made.

The lesson's learning objective is unchanged throughout: by the end, the learner can identify the key components of a condenser unit, describe what each one does in the cooling process, and locate the outdoor disconnect for safe maintenance. Every model discussed below is traced back to how it served that objective.

1. Design Process: Objective-First (Backward) Design

The lesson was built using a backward design process, the sequence popularized in Wiggins and McTighe's Understanding by Design: define the desired result first, decide what evidence would prove the learner reached it, and only then design the learning experience itself. Content and visuals were the last thing decided, not the first.

Step 1 , Desired result: The three-part learning objective (identify, describe, locate) was fixed before any script or visual existed. It appears verbatim on the opening slide and again on the closing slide, functioning as the fixed target the rest of the deck was built toward.

Step 2 , Acceptable evidence: The practice section (label the diagram, then check it against the answer key) was conceived as the proof that the objective's "identify" component had been met , a direct, gradable demonstration rather than a vague sense of exposure to the content.

Step 3 , Learning experience: The script and storyboard were then built specifically to prepare the learner for that evidence: every component that appears in the practice diagram was taught individually beforehand, in the same visual style, so nothing on the test slide is unfamiliar.

2. Structural Framework: Gagné's Nine Events of Instruction

Robert Gagné's Nine Events of Instruction describe a sequence of instructional functions , not necessarily nine separate slides, but nine jobs a lesson needs to do, in roughly this order, to move a learner from novice to competent. The storyboard's phase structure (hook, map, components, safety, review, practice, close) lines up with these events almost one-to-one:

Gagné's Event	Where It Happens	What It Does
1. Gain attention	Slides 1–2	Curiosity hook: a familiar object the learner has seen but can't explain.
2. Inform learner of objective	Slide 1	The full learning objective is stated in plain language before any content.
3. Stimulate recall of prior learning	Slide 2	Anchors the lesson to something the

Gagné's Event	Where It Happens	What It Does
		learner has already seen firsthand.
4. Present the content	Slides 3–10	The system map and the six components are taught in physical cause-and-effect order.
5. Provide learning guidance	Slides 4–9	Metaphors (“exhale,” “battery boost”) and color-coding give the learner a way to interpret new information, not just receive it.
6. Elicit performance	Slide 15	The learner labels the diagram from memory before seeing any answer.
7. Provide feedback	Slide 16	The answer key is shown immediately after, in the same layout, for direct comparison.
8. Assess performance	Slides 15–16 together	The blank-then-labeled pair functions as an embedded, self-graded check.
9. Enhance retention & transfer	Slides 12–13, 17	Recap, a second dedicated safety pass, and the closing restatement of the objective.

3. Cognitive Level Alignment: Bloom's Taxonomy

The three verbs in the learning objective were not chosen loosely , each sits at a different, deliberately ascending level of Bloom's Taxonomy of cognitive levels, and the storyboard's visual treatment escalates alongside them.

Objective Verb	Bloom's Level	How the Storyboard Serves It
“Identify the key components”	Remember	Repeated exposure to the same labeled diagram (Slides 3, 12, 15–16) builds simple recognition and recall.
“Describe what each one does”	Understand	Function-focused narration and visual metaphors on each component slide (4–9) build explanatory understanding, not just naming.
“Locate the outdoor disconnect for safe maintenance”	Apply	A numbered, physically-performed procedure (Slide 10, reinforced in Slide 13) asks the learner to act, not just recall or explain.

The visual intensity of the lesson tracks this same climb: components at the Remember/Understand level use a calm, muted palette, while the single Apply-level item , the disconnect , is the only content rendered in warning red. Higher cognitive demand and higher visual salience rise together.

4. Managing Mental Effort: Cognitive Load Theory

John Sweller's Cognitive Load Theory holds that working memory is limited and instruction should manage three kinds of load: intrinsic (the inherent difficulty of the material), extraneous (load created by poor design, not the content itself), and germane (effort that actually builds understanding). The storyboard manages all three deliberately:

Reducing extraneous load: One idea per slide, a consistently muted default palette, and clean, uncluttered hook slides (design notes for Slide 2 explicitly call for avoiding anything that isn't essential to the single point being made) all keep the learner's attention on content rather than on decoding a busy layout.

Managing intrinsic load: The six components are sequenced from the structurally simplest (Housing, a 2-second beat with no moving parts) to the most conceptually abstract (Defrost Board, a feedback-control loop), rather than presenting all six with equal weight regardless of difficulty.

Supporting germane load: Color-coding (the coil's hot-to-cool gradient) and consistent metaphors (“exhale,” “battery boost”) are germane-load tools: they cost the learner very little extra processing but actively help build the underlying mental model of the system.

Explicit phase breaks: The “Revise” and “Practice” divider slides exist purely to signal a change in cognitive mode (learn → review → test) so the learner doesn't have to infer the shift from content alone , a small extraneous-load cost avoided at zero content cost.

5. Multimedia Design: Mayer's Cognitive Theory of Multimedia Learning

Richard Mayer's Cognitive Theory of Multimedia Learning builds on Cognitive Load Theory and Allan Paivio's Dual Coding Theory, which holds that people process visual and verbal information through two separate channels that, used together, build stronger understanding than either alone. Mayer translated this into a set of concrete design principles, several of which are directly visible in the script and storyboard:

Principle	What It Means	Where It's Applied
Multimedia principle	People learn better from words + pictures than words alone.	Every slide pairs spoken narration with a purpose-built visual rather than relying on text alone.
Modality principle	Narration (spoken) + pictures beats on-screen text + pictures, since it avoids overloading the visual channel.	Explanation is carried by audio narration; on-screen text is kept to short captions and labels, not full sentences competing with the image for visual attention.
Redundancy principle	Narration and on-screen text should not repeat each other verbatim.	Storyboard captions paraphrase the narration rather than transcribing it , e.g., Slide 2's on-screen caption is a shortened restatement of the fuller spoken line, not a duplicate.
Signaling principle	Cueing essential material (color, arrows, highlighting) helps learners find what matters.	Directional air arrows, the coil's red-to-orange heat gradient, and the disconnect's warning-red banners all cue exactly what the learner should attend to.
Temporal contiguity	Corresponding words and images should appear at the same time, not sequentially.	Design notes for Slide 10/12 specify syncing each component highlight to the exact word in the narration that names it.
Spatial contiguity	Labels should sit next to what they describe, not in a separate legend.	Every diagram places its labels directly beside the relevant part (fan, coil, contactor, etc.) rather than in a numbered key off to the side.
Segmenting principle	Complex content is easier to learn broken into learner-paced chunks.	Six components are taught as six separate slides instead of one dense, fully-annotated cutaway diagram.
Coherence principle	Extraneous detail that doesn't serve the objective should be left out.	The Capacitor slide's design notes explicitly reject a technical voltage/waveform diagram in favor of a single simple spike-and-arrow animation.

6. Building the Mental Model: Advance Organizers & Concrete-to-Abstract Sequencing

Ausubel's Advance Organizers

David Ausubel argued that new information is retained far better when the learner is first given a broad, general framework to “hang” the details on, rather than encountering isolated facts and building the framework themselves after the fact. Slide 3's system map , unlabeled outlines of every part, shown performing the condenser's single job of releasing heat , is exactly this kind of organizer. It is deliberately built to be reused: the same base image returns fully labeled at the recap (Slide 12) and again as the practice exercise (Slides 15–16), so the organizer introduced at the start becomes the structure the learner fills in by the end.

Bruner's Concrete-to-Abstract Sequencing

Jerome Bruner's work on modes of representation describes a progression from enactive (action-based), to iconic (image-based), to symbolic (abstract/schematic) understanding. The storyboard follows this arc visually: it opens on a richly rendered, near-photographic illustration of a real unit (Slide 1), moves into flat iconic cutaway diagrams (Slides 3–9), and only in the electrical slides (Contactor, Capacitor, Defrost Board) introduces more symbolic, schematic representations , switches, pulses, circuit-board icons , for concepts that have no simple physical appearance. The same logic scaffolds the order of the six components themselves, starting with the purely structural Housing before asking the learner to track moving air, heat transfer, and electrical switching.

7. Motivational Design: Keller's ARCS Model

John Keller's ARCS model identifies four conditions instruction needs to sustain motivation: Attention, Relevance, Confidence, and Satisfaction. The lesson addresses all four without ever stating them explicitly:

Attention: The opening hook (Slide 2) is a deliberate curiosity gap , an object the learner has seen but can't explain , rather than a dry statement of topic.

Relevance: The lesson is framed around a real backyard scene and a genuinely useful, physically consequential skill (locating the disconnect), rather than abstract HVAC theory.

Confidence: Content is delivered in short, single-idea chunks (as short as a 2-second beat for Housing) with a clear payoff at the recap, so the learner experiences steady, visible progress rather than being confronted with the whole system's complexity at once.

Satisfaction: The closing slide restates the objective with the six component names bolded as a self-check checklist, giving the learner a concrete, immediate sense of “I can now do exactly what I was told I'd be able to do.”

8. Locking In Learning: Retrieval Practice & Spaced Repetition

The Testing Effect (Roediger & Karpicke)

Research by Henry Roediger and Jeffrey Karpicke found that actively retrieving information from memory strengthens learning more than simply re-reading or re-viewing it , commonly called the testing effect. The practice section applies this directly: Slide 15 requires the learner to recall and supply every label from memory before Slide 16 reveals the answer key, rather than presenting the completed, labeled diagram first and asking the learner to “study” it. The order of those two slides is the entire mechanism , reversing it would turn a retrieval-practice activity back into passive review.

Spaced / Distributed Repetition

Distributed-practice research consistently shows that information encountered more than once, spread across a learning session, is retained better than information seen only once. The outdoor disconnect is the only piece of content in the entire lesson that receives this treatment: it is previewed in the opening hook (Slide 2, implicitly, via the objective), taught in full with a numbered procedure (Slide 10), revisited in a dedicated second recap (Slide 13), and tested in the practice diagram (Slides 15–16). Every other component appears once. That asymmetry is intentional , spaced repetition is reserved for the single objective item with real physical safety consequences if forgotten.

9. Synthesis: Model-to-Evidence Map

A single reference table tying every model discussed above back to its clearest evidence in the finished work:

Model / Theory	Core Idea	Primary Evidence
Backward Design (Wiggins & McTighe)	Define the objective and its evidence before designing content.	Practice slides designed as evidence before the teaching slides were built.
Gagné's Nine Events	Instruction should perform nine functions in sequence.	Full lesson structure, Slides 1–17.
Bloom's Taxonomy	Objectives sit at different cognitive levels.	Identify / Describe / Locate mapped to Remember / Understand / Apply.
Cognitive Load Theory (Sweller)	Manage intrinsic, extraneous, and germane load.	One idea per slide; easy-to-hard component order; phase dividers.
Multimedia Learning (Mayer) / Dual Coding (Paivio)	Pair words and pictures through two channels without overloading either.	Narration + visual pairing; signaling via color and arrows; paraphrased captions.
Advance Organizers (Ausubel)	Give a general framework before the details.	Slide 3 system map, reused at Slides 12 and 15–16.
Concrete-to-Abstract Sequencing (Bruner)	Move from real/physical to iconic to symbolic representation.	Slide 1 realistic art → flat diagrams → electrical schematics.
ARCS Model (Keller)	Sustain motivation via attention, relevance, confidence, satisfaction.	Curiosity hook; real-world framing; short chunks; closing self-check.
Testing Effect (Roediger & Karpicke)	Retrieval strengthens memory more than review.	Blank diagram (Slide 15) precedes the answer key (Slide 16).
Spaced Repetition	Distributed exposure beats a single exposure.	Outdoor disconnect repeated across Slides 2, 10, 13, 15–16.

10. How These Models Work Together

No single theory drives this lesson , each one answers a different design question, and together they cover the full arc from planning to memory. Backward design and Gagné's Nine Events shaped the macro-structure: what phase comes before what, and why. Bloom's Taxonomy justified the specific order and visual weighting of the three objective verbs. Cognitive Load Theory and Mayer's multimedia principles shaped every individual slide's density, pacing, and pairing of narration to image. Ausubel and Bruner shaped how new information was introduced and organized so it would attach to something the learner already had. Keller's ARCS model shaped tone and framing so the learner stayed engaged throughout. And the testing effect and spaced repetition shaped the final third of the lesson, ensuring the content didn't just get taught once and disappear.

The result is a lesson where almost every visible design choice , slide order, color use, animation pacing, what gets repeated and what doesn't , traces back to a specific, named instructional principle rather than aesthetic preference alone.

A Note on Scope

This is a single micro-lesson, not a full course, so the models applied here are the ones relevant at that scale: lesson-level sequencing, multimedia design, and retention mechanics. Course-level concerns , formal learner analysis, a full assessment blueprint, or curriculum-wide spacing across multiple sessions , sit outside what a single five-minute lesson is built to do, and are not claimed here.